



Robotic Systems I

Lecture 9: Contacts and Optimization-based Control

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Template made by Panagiotis Papagiannopoulos



Manipulator Equation Reminder:

$$\underbrace{M(\mathbf{q})}_{\text{"Mass Matrix"}} \dot{\mathbf{v}} + \underbrace{C(\mathbf{q}, \mathbf{v})}_{\text{"Coriolis/Gravity Forces"}} = \underbrace{\mathbf{u}}_{\text{"Usually } \boldsymbol{\tau}} + \underbrace{\mathbf{F}_{\text{ext}}}_{\text{"External forces"}}$$

Velocity Kinematics:

$$\dot{\mathbf{q}} = \mathbf{G}(\mathbf{q}) \mathbf{v}$$

Forward Dynamics:

$$\dot{\mathbf{v}} = \mathbf{M}^{-1}(\mathbf{q}) \left(\mathbf{u} + \mathbf{F}_{\text{ext}} - \mathbf{C}(\mathbf{q}, \mathbf{v}) \right)$$

Inverse Dynamics:

$$\boldsymbol{\tau} = \mathbf{M}(\mathbf{q}) \dot{\mathbf{v}} + \mathbf{C}(\mathbf{q}, \mathbf{v}) - \mathbf{F}_{\text{ext}}$$

Manipulator Equation (again!):

$$\mathbf{M}(\mathbf{q})\dot{\mathbf{v}} + \mathbf{C}(\mathbf{q}, \mathbf{v}) = \mathbf{u} + \mathbf{F}_{\text{ext}}$$

Let's see some dimensions:

- $\mathbf{M} \in \mathbb{R}$

Manipulator Equation (again!):

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Let's see some dimensions:

- $\mathbf{M} \in \mathbb{R}^{n \times n}$
- $\mathbf{C} \in \mathbb{R}$

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- $n = n_{\text{base}} + n_{\text{joints}}$

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Manipulator Equation (floating-base):

$$\mathbf{M}(\mathbf{q})\dot{\mathbf{v}} + \mathbf{C}(\mathbf{q}, \mathbf{v}) = \begin{bmatrix} \mathbf{0}_6 \\ \boldsymbol{\tau} \end{bmatrix} + \mathbf{F}_{\text{ext}}$$

Manipulator Equation (floating-base):

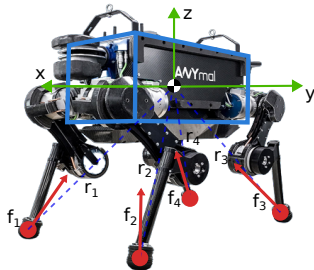
$$M(q)\dot{v} + C(q, v) = \begin{bmatrix} 0_6 \\ \tau \end{bmatrix} + F_{\text{ext}}$$

- We have **NO** direct control over the base!
- How can we control the robot?

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- We have **NO** direct control over the base!
- How can we control the robot?
- Through contact forces: F_{ext} !
- $F_{\text{ext}} = \sum_{i=1}^{N_{\text{contact}}} J_i(q)^T f_i$
- $f_i \in \mathbb{R}^3$



We assume that:

- $\dot{\mathbf{v}} \approx 0$
- $\mathbf{C}(\mathbf{q}, \mathbf{v}) \approx \mathbf{g}(\mathbf{q})$

The manipulator equation becomes:

$$\mathbf{g}(\mathbf{q}) = \begin{bmatrix} \mathbf{0}_6 \\ \boldsymbol{\tau} \end{bmatrix} + \sum_{i=1}^{N_{\text{contact}}} \mathbf{J}_i(\mathbf{q})^T \mathbf{f}_i$$

Monopod Balancing

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- **We have a robot with one leg!**

How can we find the torques $\boldsymbol{\tau}$ to apply such that we *cancel the gravity*?

- $$\begin{bmatrix} \mathbf{g}_u(\mathbf{q}) \\ \mathbf{g}_a(\mathbf{q}) \end{bmatrix} = \begin{bmatrix} \mathbf{0}_6 \\ \boldsymbol{\tau} \end{bmatrix} + \begin{bmatrix} \mathbf{J}_u(\mathbf{q})^T \mathbf{f} \\ \mathbf{J}_a(\mathbf{q})^T \mathbf{f} \end{bmatrix}$$



<https://www.umass.edu/robotics/projects/staccatoe-single-legged-hopping-robot>

Monopod Balancing (2)



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- Thus,
$$\tau = \mathbf{g}_a(\mathbf{q}) - \mathbf{J}_a(\mathbf{q})^T \mathbf{f}$$

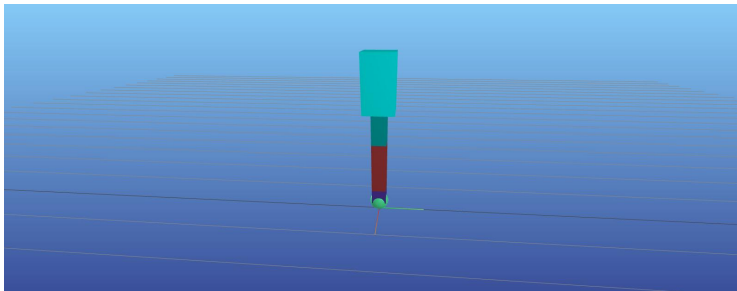
Monopod Balancing (2)



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- $$\begin{bmatrix} \mathbf{g}_u(\mathbf{q}) \\ \mathbf{g}_a(\mathbf{q}) \end{bmatrix} = \begin{bmatrix} \mathbf{0}_6 \\ \tau \end{bmatrix} + \begin{bmatrix} \mathbf{J}_u(\mathbf{q})^T \mathbf{f} \\ \mathbf{J}_a(\mathbf{q})^T \mathbf{f} \end{bmatrix}$$
- $$\mathbf{g}_u(\mathbf{q}) = \mathbf{J}_u(\mathbf{q})^T \mathbf{f}$$
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- Thus,
$$\tau = \mathbf{g}_a(\mathbf{q}) - \mathbf{J}_a(\mathbf{q})^T \mathbf{f}$$
- Is $\mathbf{J}_u(\mathbf{q})^T$ always invertible?

Monopod Balancing - Code Example



We assume that:

- $\dot{\mathbf{v}} \approx 0$
- $\mathbf{C}(\mathbf{q}, \mathbf{v}) \approx \mathbf{g}(\mathbf{q})$

Biped Manipulator Equation:

$$\mathbf{g}(\mathbf{q}) = \begin{bmatrix} \mathbf{0}_6 \\ \boldsymbol{\tau} \end{bmatrix} + \mathbf{J}_l(\mathbf{q})^T \mathbf{f}_l + \mathbf{J}_r(\mathbf{q})^T \mathbf{f}_r$$

Biped Balancing

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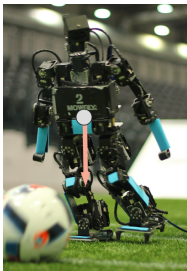
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Source

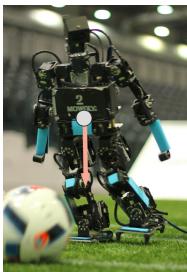
Biped Balancing (2)



Source

$$\begin{bmatrix} \mathbf{g}_u(\mathbf{q}) \\ \mathbf{g}_a(\mathbf{q}) \end{bmatrix} = \begin{bmatrix} \mathbf{0}_6 \\ \boldsymbol{\tau} \end{bmatrix} + \begin{bmatrix} (\mathbf{J}_l(\mathbf{q})^T)_u \mathbf{f}_l + (\mathbf{J}_r(\mathbf{q})^T)_u \mathbf{f}_r \\ (\mathbf{J}_l(\mathbf{q})^T)_a \mathbf{f}_l + (\mathbf{J}_r(\mathbf{q})^T)_a \mathbf{f}_r \end{bmatrix}$$

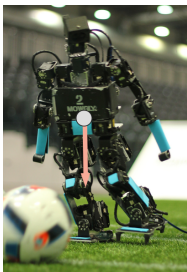
Biped Balancing (2)



Source

- $$\begin{bmatrix} \mathbf{g}_u(\mathbf{q}) \\ \mathbf{g}_a(\mathbf{q}) \end{bmatrix} = \begin{bmatrix} \mathbf{0}_6 \\ \boldsymbol{\tau} \end{bmatrix} + \begin{bmatrix} (\mathbf{J}_l(\mathbf{q})^T)_u \mathbf{f}_l + (\mathbf{J}_r(\mathbf{q})^T)_u \mathbf{f}_r \\ (\mathbf{J}_l(\mathbf{q})^T)_a \mathbf{f}_l + (\mathbf{J}_r(\mathbf{q})^T)_a \mathbf{f}_r \end{bmatrix}$$
- $$\begin{bmatrix} \mathbf{f}_l \\ \mathbf{f}_r \end{bmatrix} = [(\mathbf{J}_l(\mathbf{q})^T)_u \quad (\mathbf{J}_r(\mathbf{q})^T)_u]^\dagger \mathbf{g}_u(\mathbf{q})$$

Biped Balancing (2)



Source

- $$\begin{bmatrix} \mathbf{g}_u(\mathbf{q}) \\ \mathbf{g}_a(\mathbf{q}) \end{bmatrix} = \begin{bmatrix} \mathbf{0}_6 \\ \boldsymbol{\tau} \end{bmatrix} + \begin{bmatrix} (\mathbf{J}_l(\mathbf{q})^T)_u \mathbf{f}_l + (\mathbf{J}_r(\mathbf{q})^T)_u \mathbf{f}_r \\ (\mathbf{J}_l(\mathbf{q})^T)_a \mathbf{f}_l + (\mathbf{J}_r(\mathbf{q})^T)_a \mathbf{f}_r \end{bmatrix}$$
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- This is minimizing $\mathbf{f}_{l/r}$! We need an optimization scheme to minimize $\boldsymbol{\tau}$!
- Is it balancing?

What if we have no assumptions?

$$\mathbf{M}(\mathbf{q})\dot{\mathbf{v}} + \mathbf{C}(\mathbf{q}, \mathbf{v}) = \begin{bmatrix} \mathbf{0}_6 \\ \boldsymbol{\tau} \end{bmatrix} + \mathbf{J}(\mathbf{q})^T \mathbf{f}$$

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$$\begin{bmatrix} M_u(\mathbf{q})\dot{\mathbf{v}} \\ M_a(\mathbf{q})\dot{\mathbf{v}} \end{bmatrix} + \begin{bmatrix} \mathbf{C}_u(\mathbf{q}, \mathbf{v}) \\ \mathbf{C}_a(\mathbf{q}, \mathbf{v}) \end{bmatrix} = \begin{bmatrix} \mathbf{0}_6 \\ \boldsymbol{\tau} \end{bmatrix} + \begin{bmatrix} \mathbf{J}_u(\mathbf{q})^T \mathbf{f} \\ \mathbf{J}_a(\mathbf{q})^T \mathbf{f} \end{bmatrix}$$

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- Let's now solve for both $\dot{\mathbf{v}}$ and $\mathbf{f}$. What does this mean?

What if we have no assumptions?

$$M(q)\dot{\mathbf{v}} + \mathbf{C}(q, \mathbf{v}) = \begin{bmatrix} \mathbf{0}_6 \\ \boldsymbol{\tau} \end{bmatrix} + \mathbf{J}(q)^T \mathbf{f}$$

$$\begin{bmatrix} M_u(q)\dot{\mathbf{v}} \\ M_a(q)\dot{\mathbf{v}} \end{bmatrix} + \begin{bmatrix} \mathbf{C}_u(q, \mathbf{v}) \\ \mathbf{C}_a(q, \mathbf{v}) \end{bmatrix} = \begin{bmatrix} \mathbf{0}_6 \\ \boldsymbol{\tau} \end{bmatrix} + \begin{bmatrix} \mathbf{J}_u(q)^T \mathbf{f} \\ \mathbf{J}_a(q)^T \mathbf{f} \end{bmatrix}$$

- Let's now solve for both $\dot{\mathbf{v}}$ and $\mathbf{f}$. What does this mean?
- $\begin{bmatrix} M_u(q) & -J_u(q)^T \end{bmatrix} \begin{bmatrix} \dot{\mathbf{v}} \\ \mathbf{f} \end{bmatrix} = -\mathbf{C}_u(q, \mathbf{v})$

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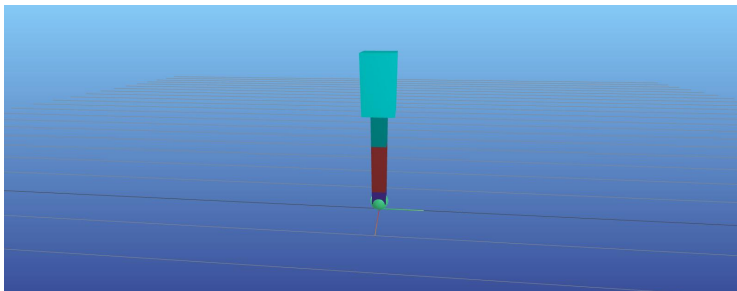
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- $\begin{bmatrix} \dot{\mathbf{v}} \\ \mathbf{f} \end{bmatrix} = -\begin{bmatrix} \mathbf{M}_u(\mathbf{q}) & -\mathbf{J}_u(\mathbf{q})^T \end{bmatrix}^\dagger \mathbf{C}_u(\mathbf{q}, \mathbf{v})$
- And $\boldsymbol{\tau} = \mathbf{M}_a(\mathbf{q})\dot{\mathbf{v}} + \mathbf{C}_a(\mathbf{q}, \mathbf{v}) - \mathbf{J}_a(\mathbf{q})^T \mathbf{f}$
- Is it balancing?

Monopod Balancing v2 - Code Example



Multi-limb Robots

Manipulator Equation (again!):

$$\mathbf{M}(\mathbf{q})\dot{\mathbf{v}} + \mathbf{C}(\mathbf{q}, \mathbf{v}) = \begin{bmatrix} \mathbf{0}_6 \\ \boldsymbol{\tau} \end{bmatrix} + \sum_{i=1}^{N_{\text{contact}}} \mathbf{J}_i(\mathbf{q})^T \mathbf{f}_i$$

- Assuming we are at state $(\mathbf{q}, \mathbf{v})$,

the dynamics are linear with respect to

$$\begin{bmatrix} \dot{\mathbf{v}} \\ \boldsymbol{\tau} \\ \mathbf{f}_1 \\ \vdots \\ \mathbf{f}_{N_{\text{contact}}} \end{bmatrix}$$

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- Why?

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- Why?

$$\begin{bmatrix} \mathbf{M}(\mathbf{q}) & -\mathbf{I}_{6+n_{\text{joints}}} & -\mathbf{J}_1(\mathbf{q})^T & \cdots & -\mathbf{J}_{N_{\text{contact}}}(\mathbf{q})^T \end{bmatrix} \begin{bmatrix} \dot{\mathbf{v}} \\ \mathbf{0}_6 \\ \boldsymbol{\tau} \\ \mathbf{f}_1 \\ \vdots \\ \mathbf{f}_{N_{\text{contact}}} \end{bmatrix} = -\mathbf{C}(\mathbf{q}, \mathbf{v})$$

Multi-limb Robots (2)

$$\begin{bmatrix} \mathbf{M}(\mathbf{q}) \\ -\mathbf{S} \\ -\mathbf{J}_1(\mathbf{q})^T \\ \vdots \\ -\mathbf{J}_{N_{\text{contact}}}(\mathbf{q})^T \end{bmatrix}^T \begin{bmatrix} \dot{\mathbf{v}} \\ \mathbf{u} \\ \mathbf{f}_1 \\ \vdots \\ \mathbf{f}_{N_{\text{contact}}} \end{bmatrix} = -\mathbf{C}(\mathbf{q}, \mathbf{v}), \text{ where } \mathbf{S} = \begin{bmatrix} \mathbf{0}_{6 \times 6} \\ \mathbf{I}_{n_{\text{joints}}} \end{bmatrix}$$

Multi-limb Robots (2)

$$\begin{bmatrix} M(\mathbf{q}) \\ -\mathbf{S} \\ -J_1(\mathbf{q})^T \\ \vdots \\ -J_{N_{\text{contact}}}(\mathbf{q})^T \end{bmatrix}^T \begin{bmatrix} \dot{\mathbf{v}} \\ \mathbf{u} \\ \mathbf{f}_1 \\ \vdots \\ \mathbf{f}_{N_{\text{contact}}} \end{bmatrix} = -\mathbf{C}(\mathbf{q}, \mathbf{v}), \text{ where } \mathbf{S} = \begin{bmatrix} \mathbf{0}_{6 \times 6} \\ I_{n_{\text{joints}}} \end{bmatrix}$$

- We set $\mathbf{x} = \begin{bmatrix} \dot{\mathbf{v}} \\ \mathbf{u} \\ \mathbf{f}_1 \\ \vdots \\ \mathbf{f}_{N_{\text{contact}}} \end{bmatrix}$ as the optimization variables and we can use

any optimizer that supports at least linear constraints!

- We usually use a QP solver:
 - Flexible enough (quadratic objectives)
 - Supports linear constraints
 - Fast enough for real-time control!

- We call this type of controllers **“Whole-Body Controllers”**
- We also refer to this type of control as **“Whole-Body Control”** (WBC)
- We can add many tasks and let the optimizer find the best solution
- We can add many constraints (*linear*)
- QPs are fast! We can also do hierarchies of solvers!
- Generic way of defining controllers

Whole Body Control via QP

$$\begin{aligned} \min_{\mathbf{x}} f(\mathbf{x}) &= \frac{1}{2} \mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{q}^T \mathbf{x} \\ \text{s.t. } \mathbf{A} \mathbf{x} - \mathbf{b} &= \mathbf{0} \\ \mathbf{C} \mathbf{x} - \mathbf{d} &\leq \mathbf{0} \end{aligned}$$

where $\mathbf{x}, \mathbf{q} \in \mathbb{R}^N$, $\mathbf{Q} \succ 0 \in \mathbb{R}^{N \times N}$.

$$\blacksquare \mathbf{x} = \begin{bmatrix} \dot{\mathbf{v}} \\ \mathbf{u} \\ \mathbf{f}_1 \\ \vdots \\ \mathbf{f}_{N_{\text{contact}}} \end{bmatrix}, \mathbf{A} = [\mathbf{M}(\mathbf{q}) \quad -\mathbf{S} \quad -\mathbf{J}_1(\mathbf{q})^T \quad \cdots \quad -\mathbf{J}_{N_{\text{contact}}}(\mathbf{q})^T],$$

$$\mathbf{b} = -\mathbf{C}(\mathbf{q}, \mathbf{v})$$

■ Multiple tasks are combined with a weighted sum: $\sum_{i=0}^{N_{\text{tasks}}} w_i \|\mathbf{W}_i \mathbf{x} - \mathbf{t}_i\|^2$

$$\blacksquare \text{ Practically, we define } \mathbf{W} = \begin{bmatrix} w_1 \mathbf{W}_1 \\ \vdots \\ w_{N_{\text{tasks}}} \mathbf{W}_{N_{\text{tasks}}} \end{bmatrix} \text{ and } \mathbf{t} = \begin{bmatrix} w_1 \mathbf{t}_1 \\ \vdots \\ w_{N_{\text{tasks}}} \mathbf{t}_{N_{\text{tasks}}} \end{bmatrix}$$

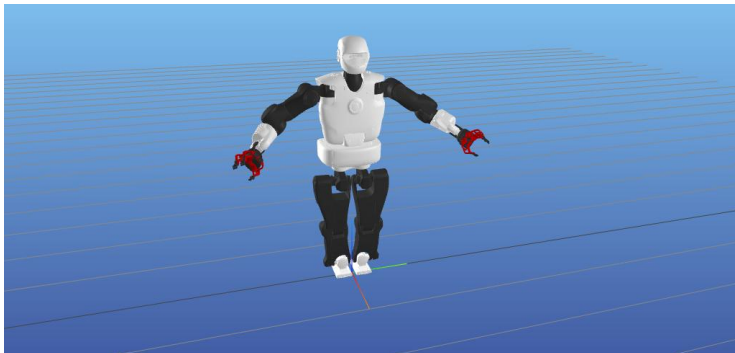
■ We set $\mathbf{Q} = \mathbf{W}^T \mathbf{W}$ and $\mathbf{q} = -\mathbf{W}^T \mathbf{t}$

■ We set $\mathbf{C}, \mathbf{d}$ according to any other constraints (like joint limits, etc.)

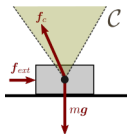
End-effector desired acceleration as a QP task:

- We have a desired acceleration $\ddot{\mathbf{p}}_d$ for an end-effector.
- $\ddot{\mathbf{p}}_d$ usually comes from a P(ID) task-space controller
- How can we make this a QP task?
- $\dot{\mathbf{p}} = \mathbf{J}(\mathbf{q})\mathbf{v} \implies \ddot{\mathbf{p}} = \dot{\mathbf{J}}(\mathbf{q})\mathbf{v} + \mathbf{J}(\mathbf{q})\dot{\mathbf{v}}$ (*product rule*)
- Thus, we get a task like $\|\mathbf{J}(\mathbf{q})\dot{\mathbf{v}} - (\ddot{\mathbf{p}}_d - \dot{\mathbf{J}}(\mathbf{q})\mathbf{v})\|^2$
- Putting this in all the optimization variables, we get $\mathbf{W}_i = \begin{bmatrix} \mathbf{J}(\mathbf{q}) & \mathbf{0} & \mathbf{0} \end{bmatrix}$ and $\mathbf{t}_i = \ddot{\mathbf{p}}_d - \dot{\mathbf{J}}(\mathbf{q})\mathbf{v}$

Whole Body Control - Code Example

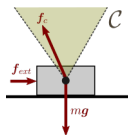


- We have been ignoring something!
Contacts do not allow for any force!
- Let's consider the 2D contact



From <https://scaron.info/>

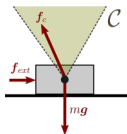
Contact Modelling



From <https://scaron.info/>

- We have been ignoring something!
Contacts do not allow for any force!
- Let's consider the 2D contact
- This contact is fixed while the force $f_c = mg - f_{ext}$ remains inside the *friction cone* $\mathcal{C}$
- We say the contact is in *fixed mode*

Contact Modelling



From <https://scaron.info/>

- We have been ignoring something!
Contacts do not allow for any force!
- Let's consider the 2D contact
- This contact is fixed while the force $\mathbf{f}_c = \mathbf{mg} - \mathbf{f}_{ext}$ remains inside the *friction cone* $\mathcal{C}$
- We say the contact is in *fixed mode*
- When $\mathbf{f}_c \notin \mathcal{C}$, it switches to the *sliding mode*
- *Rolling mode* when the body rotates over the ground/other body
- *Broken/free mode* when there is no contact force $\mathbf{f}_c$

- We usually want the contacts to be in **fixed mode**!
- Sliding contacts are hard to control
- Rolling contacts are also most of the time undesirable
- Free mode is what we already know!

Friction Cones

- For each contact we have a force being applied at the contact position: $\mathbf{f}$
- We can split the force into the *normal* component, $\mathbf{f}_n$ and the *tangential* component, $\mathbf{f}_t$
- In 3D space we have 2 tangential components: $\mathbf{f}_t$ and $\mathbf{f}_b$
- For the contact to be fixed we need to have the following:

$(\mathbf{f} \cdot \mathbf{n}) > 0$, the force cannot pull/grasp!

$$\|\mathbf{f} \cdot \mathbf{t}\|_2 \leq \mu(\mathbf{f} \cdot \mathbf{n})$$

$$\|\mathbf{f} \cdot \mathbf{b}\|_2 \leq \mu(\mathbf{f} \cdot \mathbf{n})$$

where μ is the Coulomb *static friction coefficient* at the contact, $\mathbf{n}$ is the normal direction and $\mathbf{t}, \mathbf{b}$ are the tangential directions of the contact.

Friction Cone Constraints:

$(\mathbf{f} \cdot \mathbf{n}) > 0$, the force cannot pull/grasp!

$$\|\mathbf{f} \cdot \mathbf{t}\|_2 \leq \mu(\mathbf{f} \cdot \mathbf{n})$$

$$\|\mathbf{f} \cdot \mathbf{b}\|_2 \leq \mu(\mathbf{f} \cdot \mathbf{n})$$

These are “Conic” constraints!

Friction Cones in QP

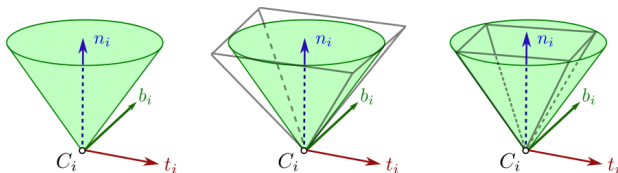
Friction Cone Constraints:

$(\mathbf{f} \cdot \mathbf{n}) > 0$, the force cannot pull/grasp!

$$\|\mathbf{f} \cdot \mathbf{t}\|_2 \leq \mu(\mathbf{f} \cdot \mathbf{n})$$

$$\|\mathbf{f} \cdot \mathbf{b}\|_2 \leq \mu(\mathbf{f} \cdot \mathbf{n})$$

These are “Conic” constraints! We need to linearize them for the QP:



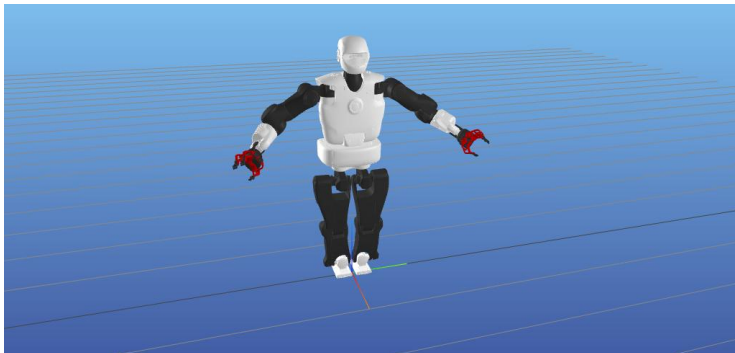
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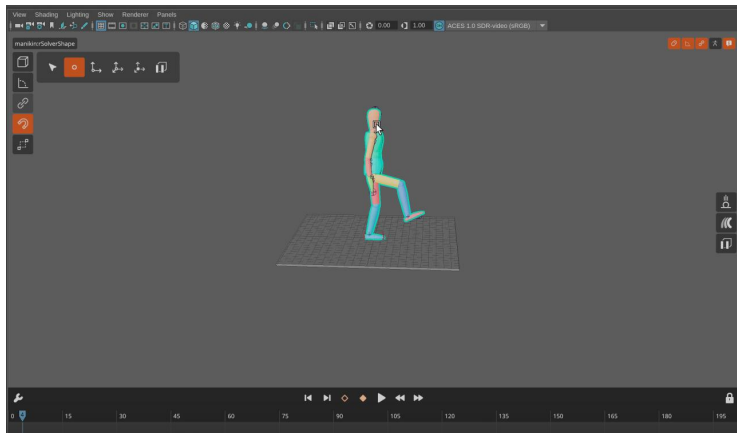
$$|\mathbf{f} \cdot \mathbf{t}| \leq \mu(\mathbf{f} \cdot \mathbf{n}) \text{ or } |\mathbf{f} \cdot \mathbf{t}| \leq \frac{\mu}{\sqrt{2}}(\mathbf{f} \cdot \mathbf{n})$$

$$|\mathbf{f} \cdot \mathbf{b}| \leq \mu(\mathbf{f} \cdot \mathbf{n}) \text{ or } |\mathbf{f} \cdot \mathbf{b}| \leq \frac{\mu}{\sqrt{2}}(\mathbf{f} \cdot \mathbf{n})$$

WBC with Friction Cones - Code Example



WBC in the wild!



Optimization Based Full Body Control for the Atlas Robot, *Feng, S., Whitman, E., Xinjilefu, X. and Atkeson, C.G.*, 2014, Humanoids. paper

Balancing experiments on a torque-controlled humanoid with hierarchical inverse dynamics, *Herzog, A., Righetti, L., Grimminger, F., Pastor, P. and Schaal, S.*, 2014, IROS. paper

Thank you

- **Any Questions?**
- **Office Hours:**
 - **Mon-Tue** (09:00-11:00)
 - 24/7 by email (costashatz@upatras.gr, subject: *ECE_RSI_AM*)
- **Material and Announcements**



Laboratory of Automation & Robotics